Tutorial 4: Confidence intervals

1. A chemical process makes a resin for use in the manufacture of plastics. One of the properties of interest is its "bulk density aerated". Forty samples are taken from the recent production and the resulting determinations of this bulk density are given below:

.341	.368	.358	.335	.347	.373	.365	.332
.330	.362	.364	.352	.338	.327	.350	.358
.385	.370	.350	.341	.349	.354	.330	.352
.377	.355	.324	.375	.338	.335	.382	.355
.338	.373	.352	.385	.349	.359	.378	.346

   For these data: $\sum x_i = 14.152$ and $\sum x_i^2 = 5.018046$.

   Determine a 95% confidence interval for the mean bulk density of the resin in the recent production.

2. A company makes sausage skins for use in the automated manufacture of sausages. A critical quantity is the 'bud length' of the crimped endpiece. In a quality control investigation a random sample of twenty bud lengths is taken from the day's production. The resulting lengths in mm are:

115	82	96	87	100
69	108	116	180	99
94	86	104	98	103
102	62	94	126	96

   with

$$\sum x_i = 2017 \qquad \text{and} \qquad \sum x_i^2 = 214253.$$

   (a) Assuming normality of bud lengths, calculate a 95% confidence interval for the mean μ of the day's production.

   (b) Plot the 20 observations informatively and comment on the normality assumption.

   (c) The observation 180mm was found to be incorrectly recorded and should have been only 80mm. Comment on the normality assumption in light of this.

   (d) Recalculate the confidence interval for the corrected data.

   (e) Use a diagram to compare the confidence intervals and comment.

3. For the data in Question 2, calculate a 95% confidence interval for the s.d. σ of the day's production, when the observation 180mm has been corrected to 80mm.

   What is the main assumption on which this confidence interval is based?

4. For a random sample of 7 vehicles travelling over a particular stretch of motorway, radar monitoring recorded the following speeds, in miles per hour (*mph*):

$$79 \quad 73 \quad 68 \quad 77 \quad 86 \quad 71 \quad 69.$$

For these data $\sum x_i = 523$ and $\sum x_i^2 = 39321$.

(a) Stating clearly any assumptions you need to make, calculate a 95% confidence interval for the unknown mean speed, μ, of all vehicles travelling over this stretch of motorway.

(b) Suppose now that the speed of 86 *mph* in the above sample was wrongly recorded, and that the correct recording is instead 76 *mph*. Without doing any calculations, state whether a 95% confidence interval for μ will now be wider or narrower than the one found in (a). Give reasons for your answer.

5. Suppose that in an employment survey we want to determine the percentage of unemployed people in a certain population, p, in a way such that our estimation has a margin of error less than 0.5% with probability 95%. In a similar study conducted a year ago it was found that the percentage of unemployed people in the population was 6%. Find the sample size, n, that is required to achieve this margin of error.

(*Hint: Consider a* 95% *CI for* p *and find* n *such that* $|p - \hat{p}| \leq 0.005$. *Also, consider the unemployment figure from the earlier study as an estimate of* p *in the expression for the standard error in the CI.*)

Exam Style Question

The data $\underline{x} = (x_1, \ldots, x_{19})$ below are the scores of 19 randomly chosen candidates in a psychometric test for an accountancy internship.

79	90	101	83	72	76	71
93	91	87	76	86	80	89
77	69	61	86	81		

Note that

$$\sum_{i=1}^{19} x_i = 1548 \quad \text{and} \quad \sum_{i=1}^{19} (x_i - \bar{x})^2 = 1670.737.$$

(a) By calculating the appropriate numerical statistics, construct a box plot for the data $\underline{x}$.

[4 marks]

(b) Explain why

$$h(\underline{X}, \mu) = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

is a pivotal quantity for the population mean, μ, of all candidates taking the test.

[3 marks]

(c) Calculate a 95% confidence interval for the population mean based on the sample results. Justify any assumptions you make in obtaining this interval.

[3 marks]

(d) The HR department claims that the average score for all candidates taking this psychometric test is 85. Does this seem a reasonable claim? Use (c) to justify your answer.

[1 mark]

[Total 11 marks]